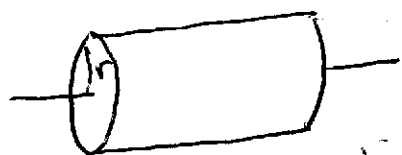


Examen Parcial Física III

Solucionario

Prob 1

56



usando ley de Gauss

$$\oint \vec{E} \cdot d\vec{l} = Q_{in}/\epsilon_0$$

$$E 2\pi r L = Q/\epsilon_0 \quad (1P)$$

$$\vec{E} = \frac{Q}{2\pi\epsilon_0 L} \frac{1}{r} \hat{e}_r \quad (1P)$$

Calculo de F $d\vec{F} = dQ \vec{E}$

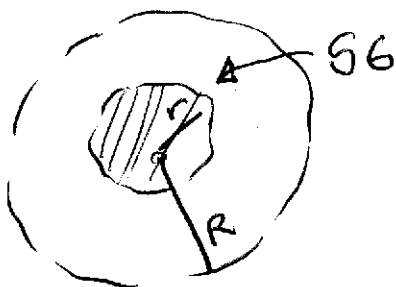
$$d\vec{F} = \left(\frac{Q}{L} dr\right) \frac{1}{2\pi\epsilon_0 r} \hat{e}_r$$

$$F = \frac{dQ}{2\pi\epsilon_0 L} \int_a^b \frac{dr}{r}$$

$$\vec{F} = \frac{dQ}{2\pi\epsilon_0 L} \ln\left(\frac{L+d}{d}\right) \hat{r} \quad (2P)$$

Prob 2:

Calculo de Q_{in} y δ



$$dQ = 34\pi r^2 dr = 248\pi r^3 dr$$

$$Q(r) = 6\pi r^4$$

$$a) \left\{ \delta = \frac{Q}{6\pi R^4} \right\} \quad Q(r) = \frac{Q}{R^4} r^4 \quad (1P)$$

b) E $r \leq R$

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q(r)}{\epsilon_0}$$

$$E 4\pi r^2 = \frac{Q}{\epsilon_0 R^4} r^4 \quad (1P)$$

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 R^4} r^2 \hat{e}_r = \frac{3}{2} \frac{\delta}{\epsilon_0} r^2 \hat{e}_r \quad \left\{ r \leq R \right.$$

Para $r > R$

$$c) \left\{ \vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{e}_r = \frac{3}{2} \frac{R^4 \delta}{\epsilon_0 r^2} \hat{e}_r \right\} \quad (1P)$$

c) Calculo $V(r)$ $R < r$

$$V(r) - V_{\infty} = - \int_{\infty}^r \vec{E} \cdot d\vec{l}$$

$$V(r) = \frac{Q}{4\pi\epsilon_0 r} = \frac{3R^4 \delta}{2\epsilon_0 r} \quad (1P)$$

$V(r)$ $R \geq r$

$$V(r) - V(R) = - \int_R^r \frac{Q}{4\pi\epsilon_0 R^4} r^2 \hat{e}_r \cdot (\hat{e}_r dr) \quad (1P)$$

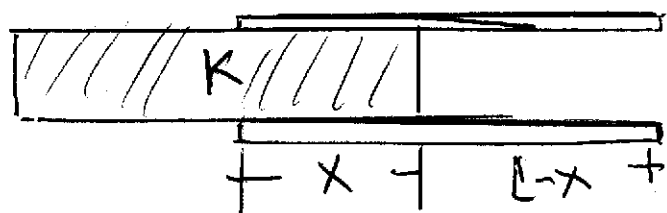
$$\left\{ \begin{aligned} V(r) &= \frac{Q}{6\pi\epsilon_0 R} \left[\frac{r^3}{2R^3} + 1 \right] \\ V(r) &= \frac{\delta R^4}{\epsilon_0} \left[\frac{r^3}{2R^3} + 1 \right] \end{aligned} \right\} \quad (1P)$$

Calculo de la Energía

$$U = \frac{1}{2} \int_0^R V dq = \frac{1}{2} \frac{Q}{6\pi\epsilon_0 R} \left[\frac{r^3}{2R^3} + 1 \right] \frac{4Q}{R^4} r^3 dr \quad (1P)$$

$$\left\{ U = \frac{3}{20} \frac{Q^2}{\pi\epsilon_0 R} = \frac{27\pi \times 10^{-2} R^7}{7\epsilon_0} \right\} \quad (1P)$$

Prob 3 : del Sistema de Condensadores



$$C_1 = \frac{K \epsilon_0 x L_y}{d} \quad C_2 = \frac{\epsilon_0 (L-x) L_y}{d} \quad (1P)$$

$$V_0 = \epsilon_0$$

$$U = \frac{1}{2} C_1 V_0^2 + \frac{1}{2} C_2 V_0^2$$

reemplazando

$$U = \frac{1}{2} \frac{\epsilon_0 L_y V_0^2}{d} [L + x(K-1)] \quad (2P)$$

$$\left\{ F = -\frac{\partial U}{\partial x} = -\frac{1}{2} \frac{\epsilon_0 L_y V_0^2}{d} (K-1) \right\} \quad (1P)$$

como $U_0 = \frac{1}{2} \frac{\epsilon_0 L L_y V_0^2}{d}$

$$\frac{U}{U_0} = 1 + \frac{x}{L} (K-1) > 0$$

(i) $U > U_0$ la Energía se incrementa (1P)

Prob 4:

$$c) \epsilon = K \epsilon_0 = 2,21 \times 10^{-11} \frac{C^2}{Nm^2} \quad (\frac{1}{2}P)$$

$$(c) \Delta V = E d$$

$$\Delta V = (2 \times 10^7)(4 \times 10^{-3}) \quad (\frac{1}{2}P)$$

$$\Delta V = 80 \text{ kV}$$

$$(cc) \sigma_0 = E K \epsilon_0$$

$$\sigma_0 = (2 \times 10^7)(2,21 \times 10^{-11}) \quad (1P)$$

$$\sigma_0 = 4,42 \times 10^{-4} \text{ C/m}^2$$

$$(iv) \sigma_i = \sigma_0 \left(1 - \frac{1}{K}\right) \quad (1P)$$

$$\sigma_i = 2,65 \times 10^{-4} \frac{C}{m^2}$$